

Machine Learning for Data Science (CS4786)

Lecture 3

Principal Component Analysis

DIM REDUCTION: LINEAR TRANSFORMATION

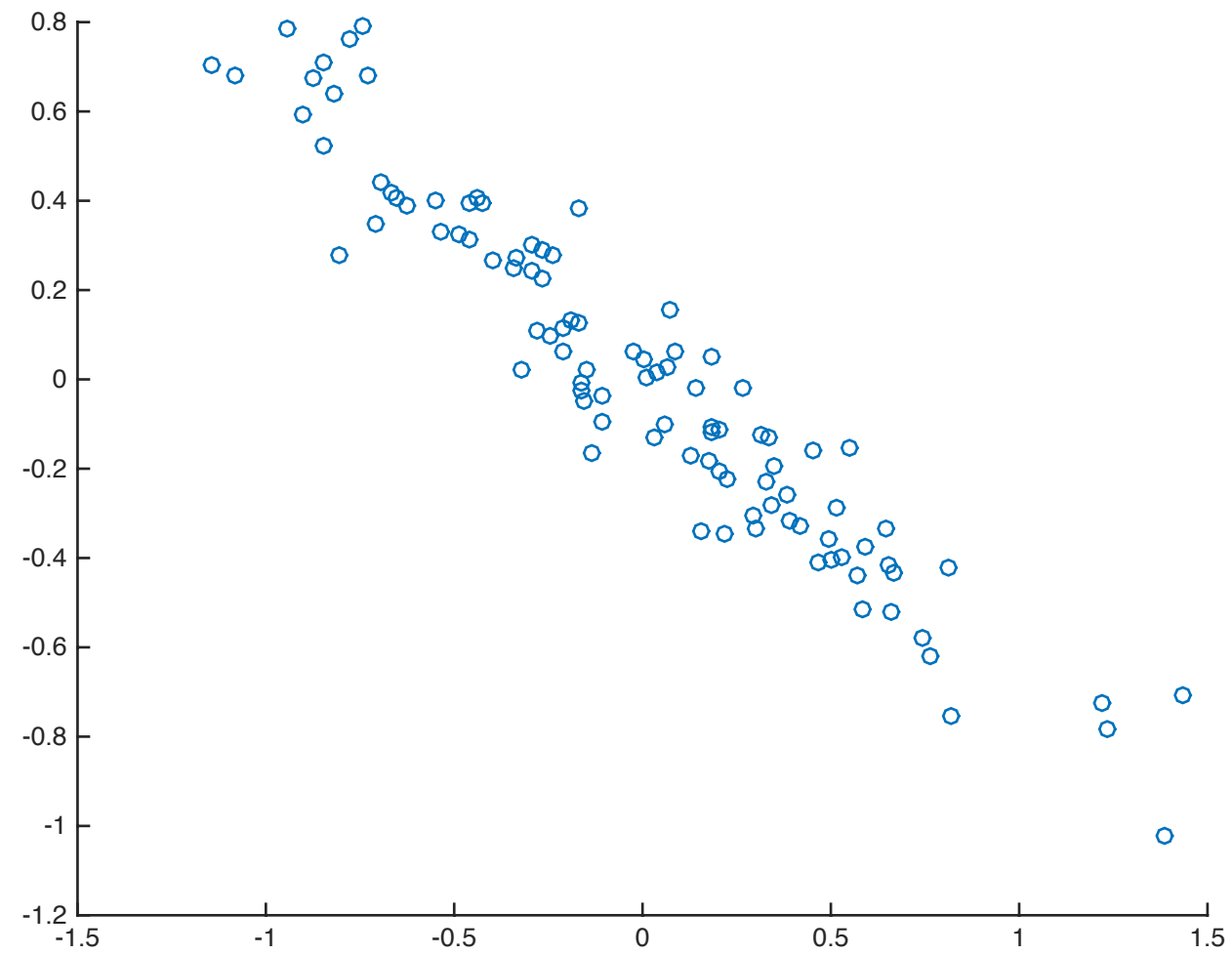
$$\begin{matrix} n \\ \times \\ d \end{matrix} X \begin{matrix} \times \\ d \\ K \end{matrix} W = \begin{matrix} n \\ \times \\ K \end{matrix} Y$$

$y_i^\top = \mathbf{x}_i^\top W$

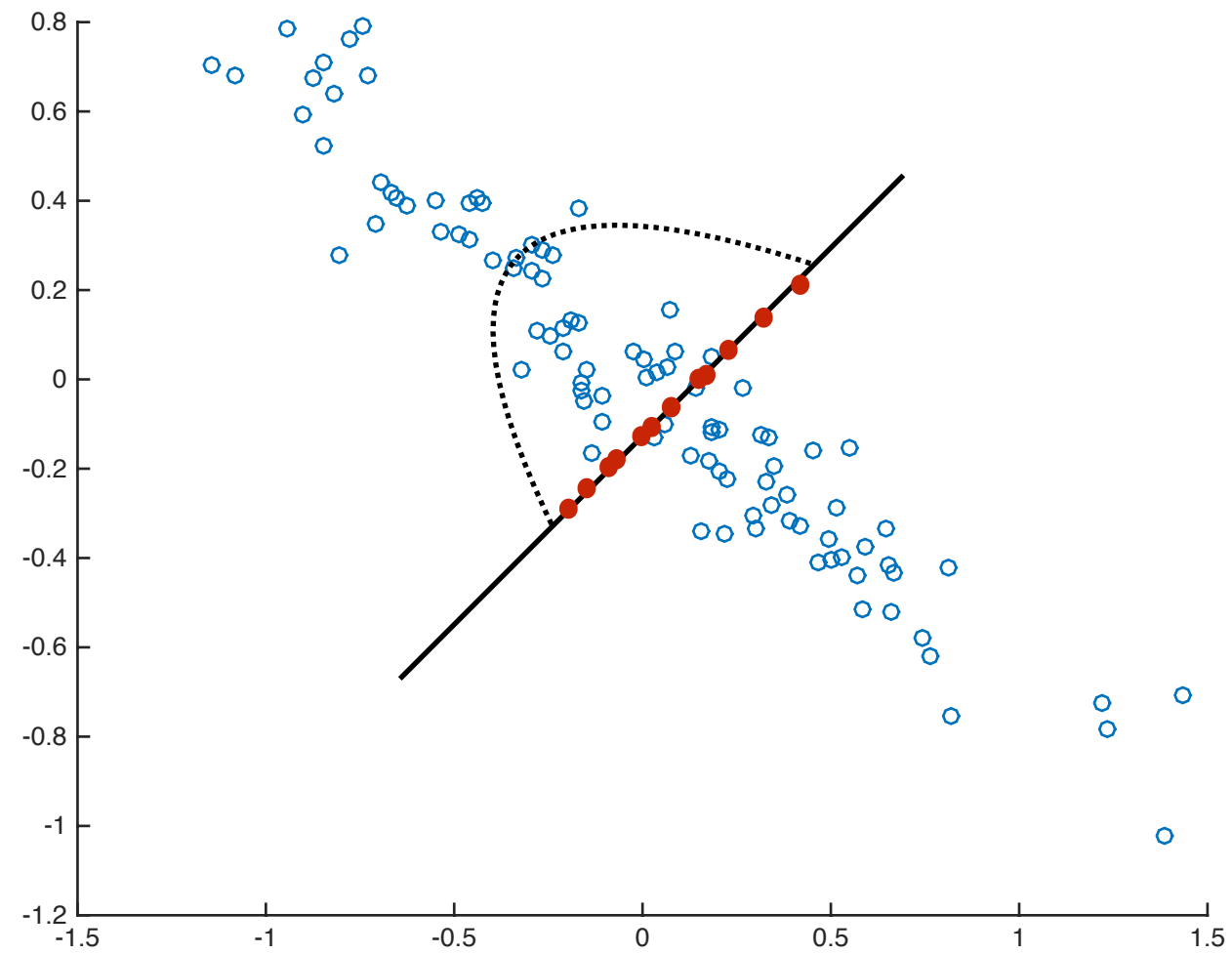
Prelude: Reducing to 1 Dim

- W is a $d \times 1$ matrix (d dimensional vector)
- Each data point is compressed to a single number
- How do we pick this W ?

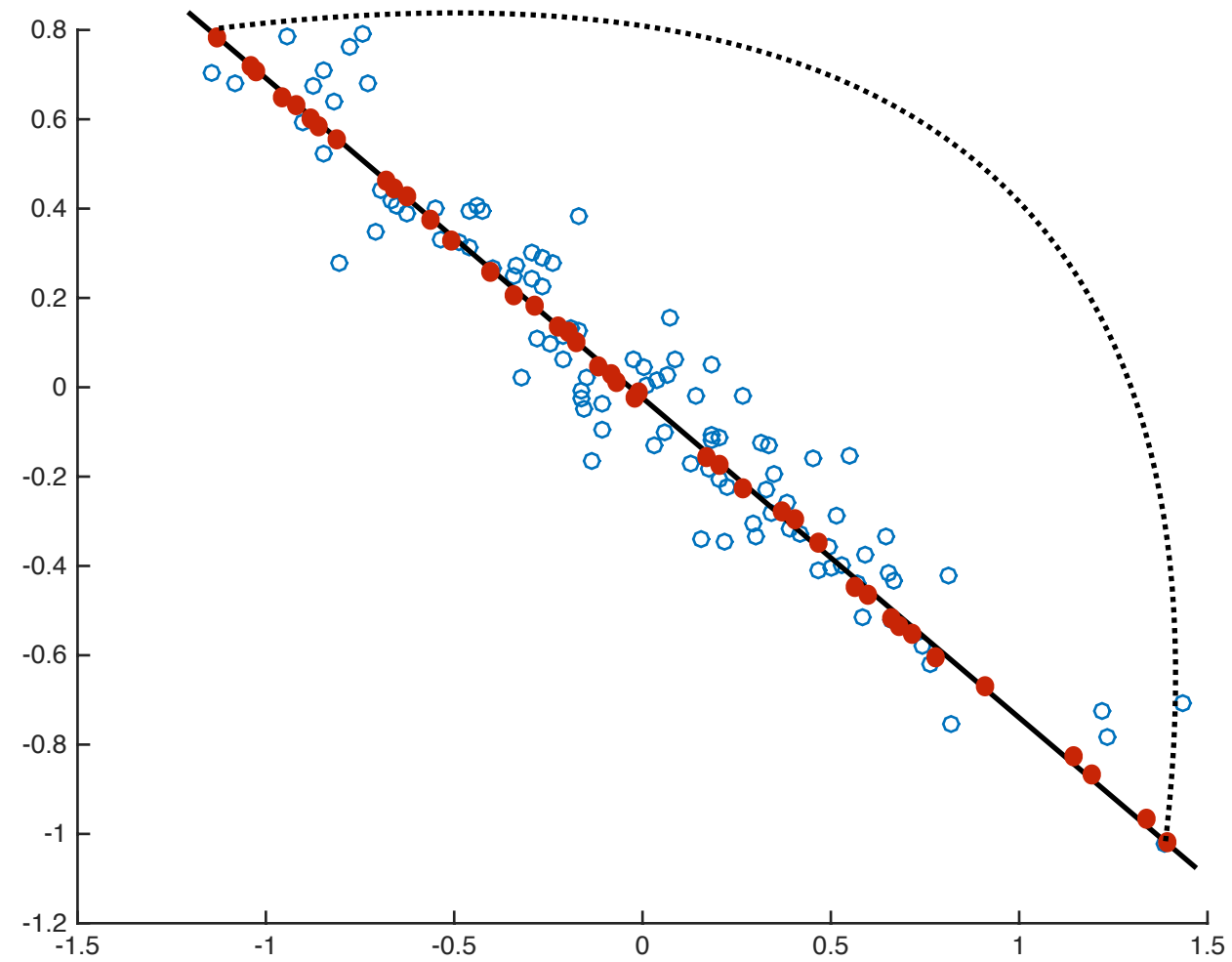
PCA: VARIANCE MAXIMIZATION



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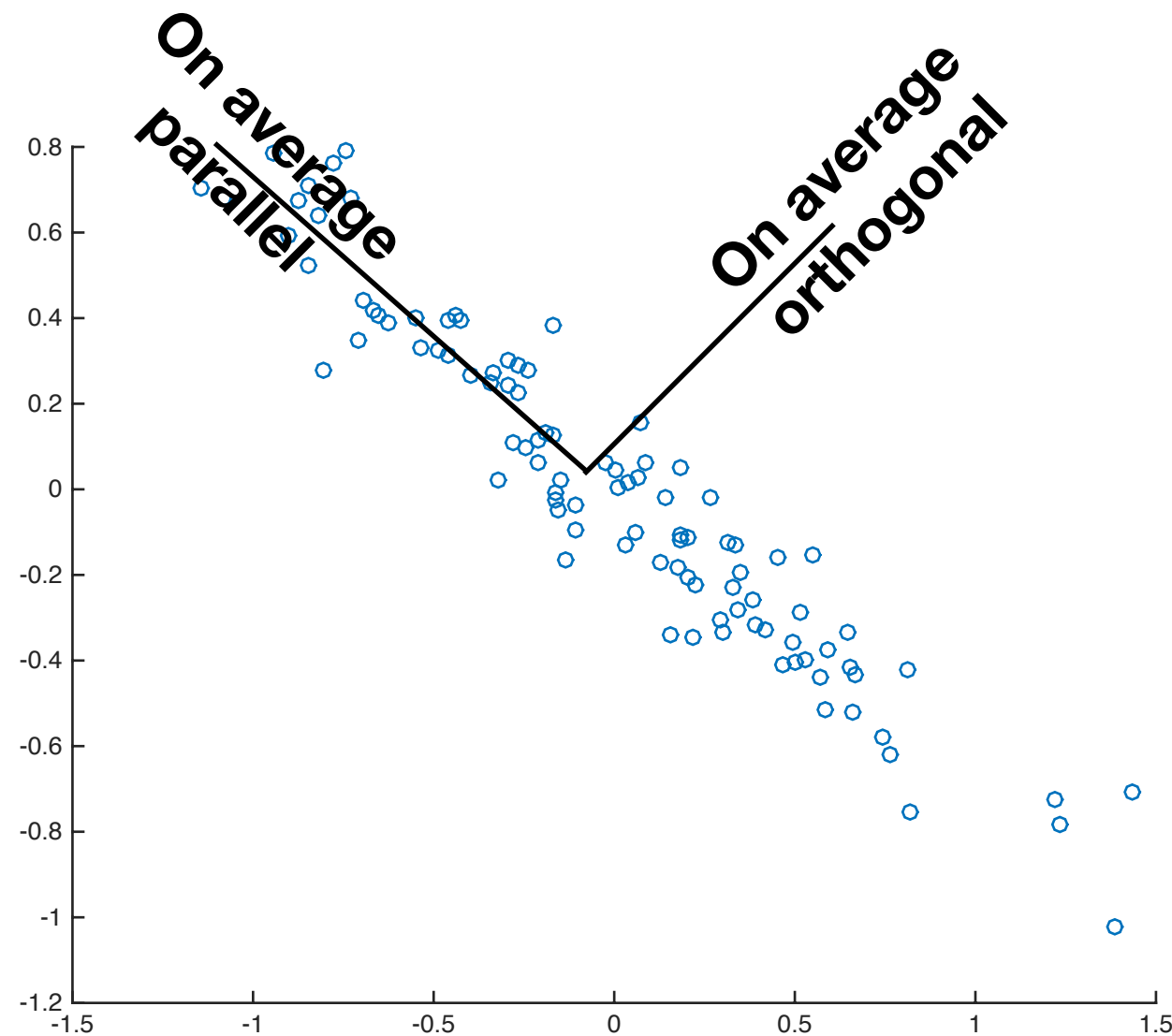


PCA: VARIANCE MAXIMIZATION

- Pick directions along which data varies the most

$$\begin{aligned}\text{Variance} &= \frac{1}{n} \sum_{t=1}^n \left(y_t - \frac{1}{n} \sum_{s=1}^n y_s \right)^2 \\ &= \frac{1}{n} \sum_{t=1}^n \left(\mathbf{w}^\top \mathbf{x}_t - \frac{1}{n} \sum_{s=1}^n \mathbf{w}^\top \mathbf{x}_s \right)^2 \\ &= \frac{1}{n} \sum_{t=1}^n \left(\mathbf{w}^\top \mathbf{x}_t - \mathbf{w}^\top \left(\frac{1}{n} \sum_{s=1}^n \mathbf{x}_s \right) \right)^2 \\ &= \frac{1}{n} \sum_{t=1}^n \left(\mathbf{w}^\top (\mathbf{x}_t - \mu) \right)^2 \\ &= \text{average squared inner product}\end{aligned}$$

Which Direction?



$$\frac{1}{n} \sum_{t=1}^n (\mathbf{w}^\top (\mathbf{x}_t - \mu))^2 = \frac{1}{n} \sum_{t=1}^n \|\mathbf{x}_t - \mu\|^2 \cos^2(w, \mathbf{x}_t - \mu)$$

PCA: VARIANCE MAXIMIZATION

- Pick directions along which data varies the most
- First principal component:

$$\begin{aligned}\mathbf{w}_1 &= \arg \max_{\mathbf{w}: \|\mathbf{w}\|_2=1} \frac{1}{n} \sum_{t=1}^n \left(\mathbf{w}^\top \mathbf{x}_t - \frac{1}{n} \sum_{t=1}^n \mathbf{w}^\top \mathbf{x}_t \right)^2 \\ &= \arg \max_{\mathbf{w}: \|\mathbf{w}\|_2=1} \frac{1}{n} \sum_{t=1}^n \left(\mathbf{w}^\top (\mathbf{x}_t - \mu) \right)^2 \\ &= \arg \max_{\mathbf{w}: \|\mathbf{w}\|_2=1} \frac{1}{n} \sum_{t=1}^n \mathbf{w}^\top (\mathbf{x}_t - \mu) (\mathbf{x}_t - \mu)^\top \mathbf{w} \\ &= \arg \max_{\mathbf{w}: \|\mathbf{w}\|_2=1} \mathbf{w}^\top \Sigma \mathbf{w}\end{aligned}$$

Σ is the covariance matrix

PCA: VARIANCE MAXIMIZATION

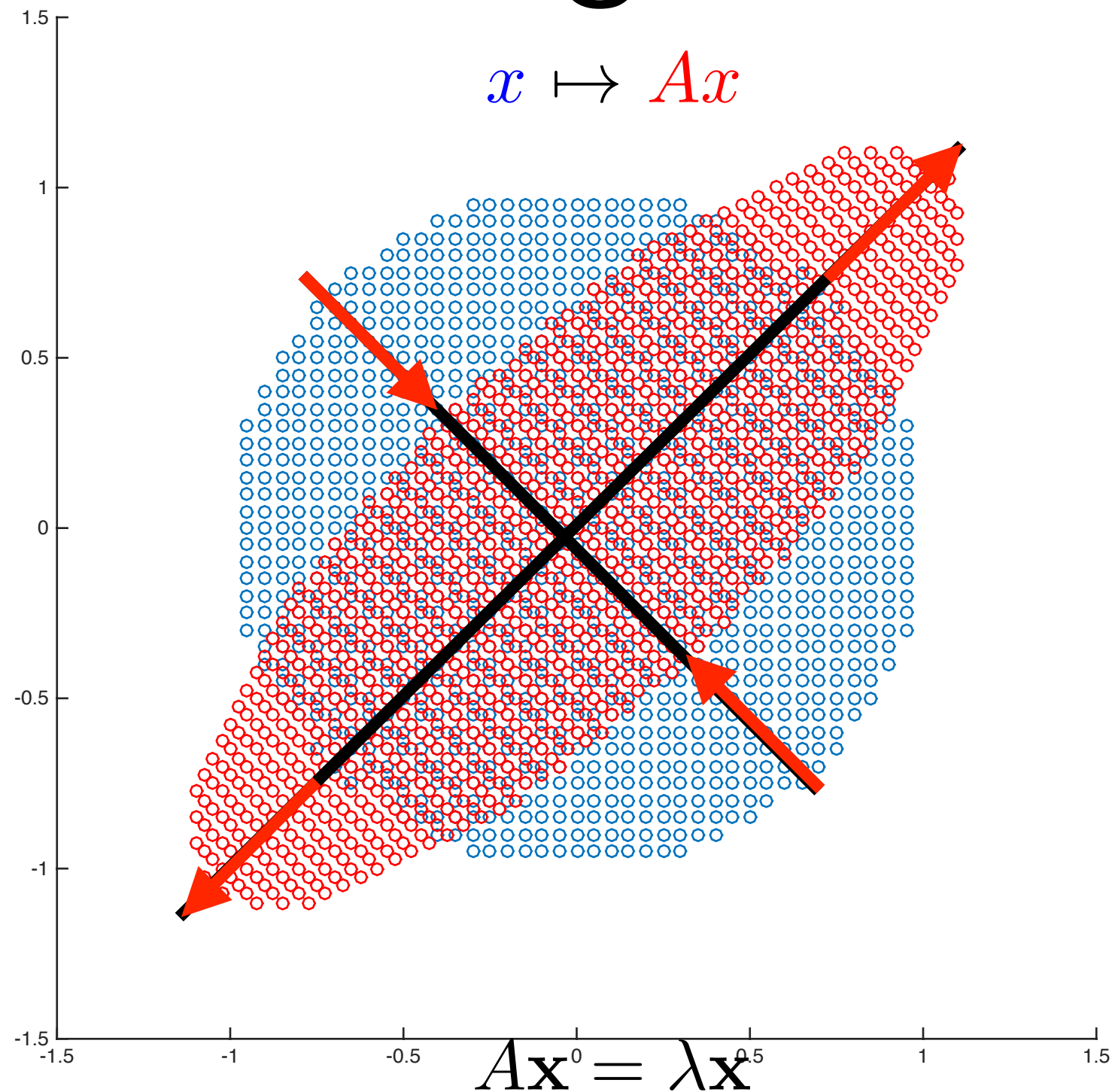
- Pick directions along which data varies the most
- First principal component:

$$\mathbf{w}_1 = \arg \max_{\mathbf{w}: \|\mathbf{w}\|_2=1} \mathbf{w}^\top \Sigma \mathbf{w}$$

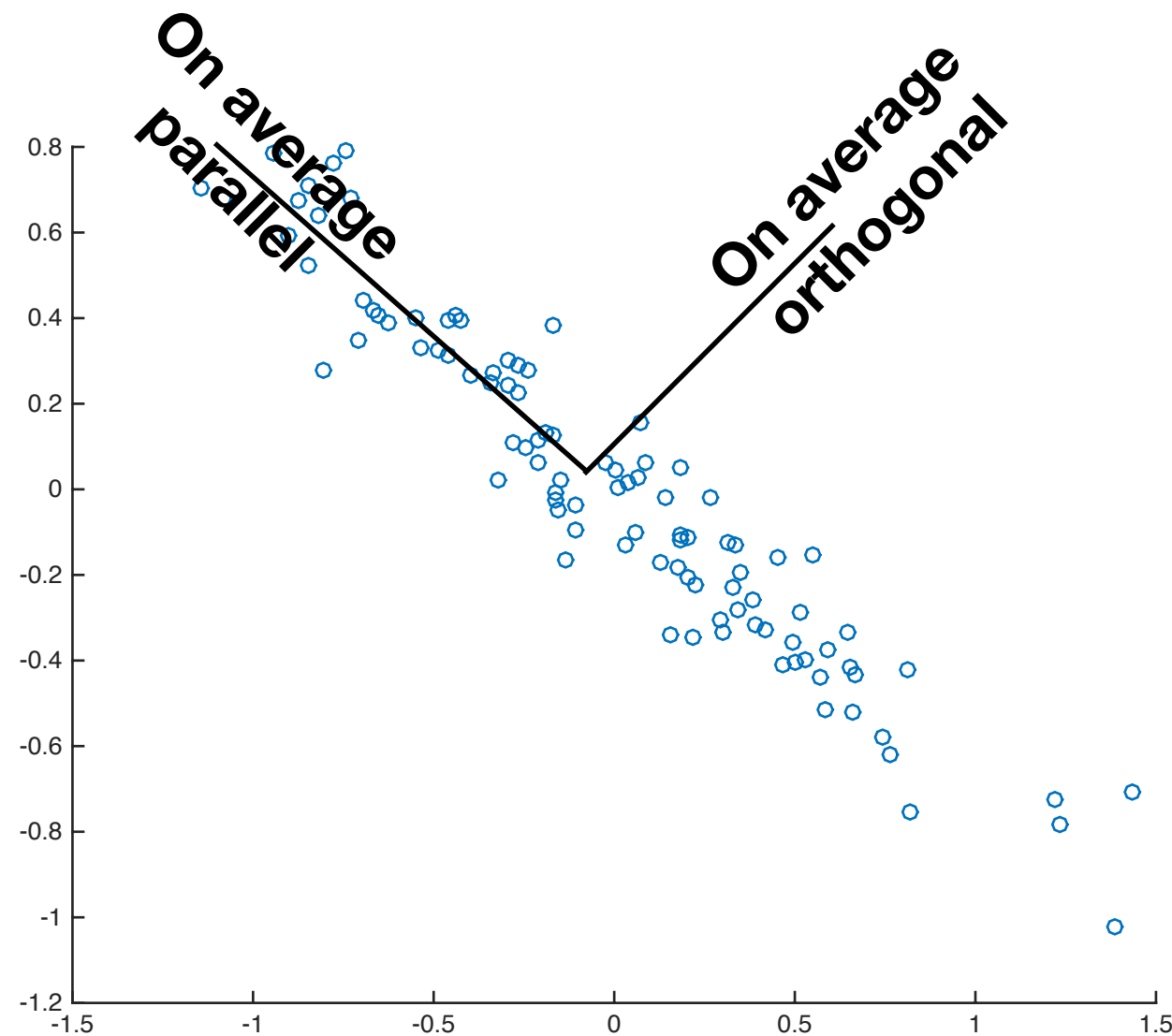
Σ is the covariance matrix

Solution: $\mathbf{w}_1 =$ Largest Eigenvector of Σ

What are Eigen Vectors?



Which Direction?



Top Eigenvector of covariance matrix

PCA: VARIANCE MAXIMIZATION

- How do we find the K components?
- We are looking for orthogonal directions that maximize total spread in each direction
- Find orthonormal W that maximizes $\sum_{k=1}^d \mathbf{w}_i[k] \mathbf{w}_j[k] = 0$ & $\sum_{k=1}^d \mathbf{w}_i[k]^2 = 1$
$$\sum_{j=1}^K \frac{1}{n} \sum_{t=1}^n \left(\mathbf{y}_t[j] - \frac{1}{n} \sum_{t=1}^n \mathbf{y}_t[j] \right)^2 = \sum_{j=1}^K \frac{1}{n} \sum_{t=1}^n \left(\mathbf{w}_j^\top \left(\mathbf{x}_t - \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t \right) \right)^2$$
$$= \sum_{j=1}^K \mathbf{w}_j^\top \Sigma \mathbf{w}_j$$

Intuition: Remove top direction, now reduce dimension for remaining d-1 dimensions

- This solutions is given by $W =$ Top K eigenvectors of Σ

PRINCIPAL COMPONENT ANALYSIS

1. $\Sigma = \text{cov}\left(X\right)$

2. $W = \text{eigs}\left(\Sigma, K\right)$

3. $Y = X \times W$

PCA: MINIMIZING RECONSTRUCTION ERROR

- Goal: find the basis that minimizes reconstruction error,

$$\begin{aligned}\|\hat{\mathbf{x}}_t - \mathbf{x}_t\|_2^2 &= \left\| \sum_{j=1}^K \mathbf{y}_t[j] \mathbf{w}_j + \mu - \mathbf{x}_t \right\|_2^2 \\&= \left\| \sum_{j=1}^K \mathbf{y}_t[j] \mathbf{w}_j + \mu - \sum_{j=1}^d \mathbf{y}_t[j] \mathbf{w}_j - \mu \right\|_2^2 \\&= \left\| \sum_{j=K+1}^d \mathbf{y}_t[j] \mathbf{w}_j \right\|_2^2 \quad (\text{but } \|a + b\|_2^2 = \|a\|_2^2 + \|b\|_2^2 + 2a^\top b) \\&= \left(\sum_{j=K+1}^d \mathbf{y}_t[j]^2 \|\mathbf{w}_j\|_2^2 + 2 \sum_{j=K+1}^d \sum_{i=j+1}^d \mathbf{y}_t[j] \mathbf{y}_t[i] \mathbf{w}_j^\top \mathbf{w}_i \right) \\&= \sum_{j=K+1}^d \mathbf{y}_t[j]^2 \|\mathbf{w}_j\|_2^2 \quad (\text{last step because } \mathbf{w}_j \perp \mathbf{w}_i)\end{aligned}$$

PCA: MINIMIZING RECONSTRUCTION ERROR

Minimize w.r.t. $\mathbf{w}_1, \dots, \mathbf{w}_K$'s that are orthonormal,

$$\begin{aligned} & \underset{\forall j, \|\mathbf{w}_j\|_2=1}{\operatorname{argmin}} \sum_{j=k+1}^d \mathbf{w}_j^\top \Sigma \mathbf{w}_j \\ &= \underset{\forall j, \|\mathbf{w}_j\|_2=1}{\operatorname{argmin}} \left(\frac{1}{n} \sum_{t=1}^n \|\hat{\mathbf{x}}_t - \mu\|_2^2 - \sum_{j=1}^K \mathbf{w}_j^\top \Sigma \mathbf{w}_j \right) \\ &= \underset{\|\mathbf{w}_j\|_2=1, \mathbf{w}_j \perp \mathbf{w}_k}{\operatorname{argmax}} \sum_{j=1}^K \mathbf{w}_j^\top \Sigma \mathbf{w}_j \end{aligned}$$